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(/learn/data-manipulation/home/welcome)

**Bill Howe**

Relational Databases are the workhouse of large-scale data management. Although originally motivated by problems in enterprise operations, they have proven remarkably capable for analytics as well. But most importantly, the principles underlying relational databases are universal in managing, manipulating, and analyzing data at scale. Even as the landscape of

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RELATIONAL DATABASES AND THE RELATIONAL ALGEBRA

Lesson 6:
Principles of
Data
Manipulation
and
Management

[Start Lesson](#) (/learn/data-manipulation/lecture/9Gufp/data-models-terminology)



Data Models, Terminology 6 min

(/learn/data-manipulation/lecture/9Gufp/data-models-terminology)



From Data Models to
Databases 5 min

(/learn/data-manipulation/lecture/uWw1n/from-data-models-to-databases)



Pre-Relational Databases 5 min

(/learn/data-manipulation/lecture/o4kbj/pre-relational-

Help Center

databases)

-  Motivating Relational Databases 3 min

(/learn/data-manipulation/lecture/FdVBF/motivating-relational-databases)

-  Relational Databases: Key Ideas 4 min

(/learn/data-manipulation/lecture/e7jWe/relational-databases-key-ideas)

Lesson 7: Relational Algebra

-  Algebraic Optimization Overview 7 min

(/learn/data-manipulation/lecture/IACV2/algebraic-optimization-overview)

-  Relational Algebra Overview 4 min

(/learn/data-manipulation/lecture/tv5TJ/relational-algebra-overview)

-  Relational Algebra Operators: Union, Difference, Selection 6 min

(/learn/data-manipulation/lecture/4JKs1/relational-algebra-operators-union-difference-selection)

-  Relational Algebra Operators: Projection, Cross Product 5 min

(/learn/data-manipulation/lecture/syjoN/relational-algebra-operators-projection-cross-product)

-  Relational Algebra Operators: Cross Product cont'd, Join 6 min

(/learn/data-manipulation/lecture/x3Lgm/relational-algebra-operators-cross-product-cont-d-join)

-  Relational Algebra Operators: Outer Join 4 min

(/learn/data-manipulation/lecture/wvpOQ/relational-algebra-operators-outer-join)

-  Relational Algebra Operators: Theta-Join 4 min

(/learn/data-manipulation/lecture/ppEF5/relational-algebra-operators-theta-join)

Lesson 8: SQL for Data Science

-  From SQL to RA 6 min

(/learn/data-manipulation/lecture/FR0Ke/from-sql-to-ra)

- ▶ Thinking in RA: Logical Query Plans 4 min

(/learn/data-manipulation/lecture/6JV6s/thinking-in-ra-logical-query-plans)

- ▶ Practical SQL: Binning Timeseries 5 min

(/learn/data-manipulation/lecture/S40H6/practical-sql-binning-timeseries)

- ▶ Practical SQL: Genomic Intervals 7 min

(/learn/data-manipulation/lecture/wiPbb/practical-sql-genomic-intervals)

- ▶ User-Defined Functions 4 min

(/learn/data-manipulation/lecture/3uBRq/user-defined-functions)

- ▶ Support for User-Defined Functions 4 min

(/learn/data-manipulation/lecture/b7Uh4/support-for-user-defined-functions)

Lesson 9: Key Principles of Relational Databases

- ▶ Optimization: Physical Query Plans 5 min

(/learn/data-manipulation/lecture/TxtOa/optimization-physical-query-plans)

- ▶ Optimization: Choosing Physical Plans 5 min

(/learn/data-manipulation/lecture/7kZQ6/optimization-choosing-physical-plans)

- ▶ Declarative Languages 5 min

(/learn/data-manipulation/lecture/NT7m3/declarative-languages)

- ▶ Declarative Languages: More Examples 5 min

(/learn/data-manipulation/lecture/wimSB/declarative-languages-more-examples)

- ▶ Views: Logical Data Independence 5 min

(/learn/data-manipulation/lecture/zjeM1/views-logical-data-

independence)



Indexes 7 min

(/learn/data-manipulation/lecture/5oK1y/indexes)

Assignment 2: SQL



Assignment: SQL for Data Science
Assignment 3h 00m

(/learn/data-manipulation/programming/nkglo/sql-for-data-science-assignment)